

- 1 For the network given in Figure 1, determine (i) I_B , (ii) I_C , (iii) β , and (iv) V_{CE}

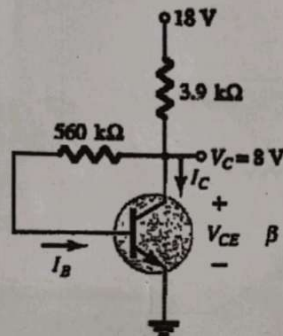


Fig. 1

- 2 For the circuit given below in Figure 2, determine- (i) I_{BQ} , (ii) I_{CQ} , (iii) V_{CEQ} and (iv) V_E .

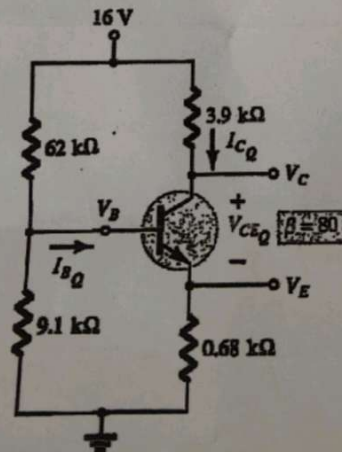


Fig. 2

- 3 Explain the working of N-Channel Enhancement type MOSFET and also draw its transfer and drain characteristics.

- 4 Determine V_2 and V_3 in below circuit shown in Figure 3.

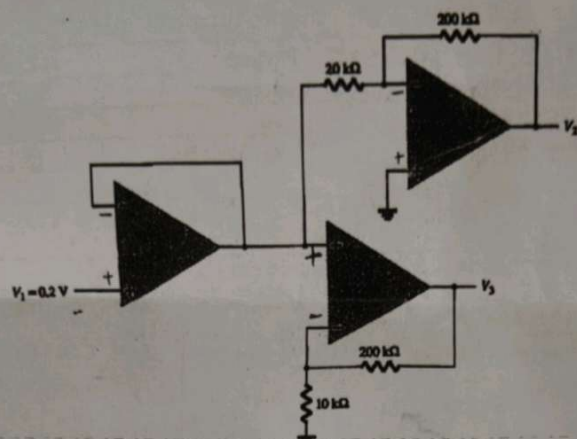


Fig. 3

- 5 Calculate the output voltage (V_o) for below circuit shown in Figure 4, given that $R_f = 330\text{K}\Omega$.

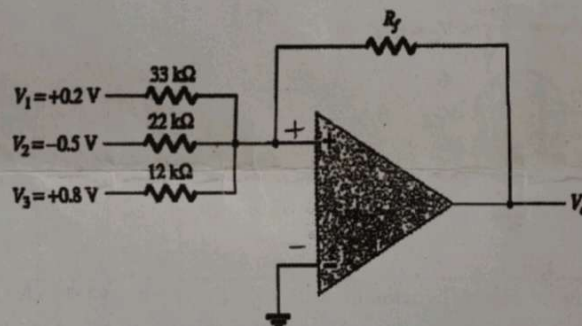
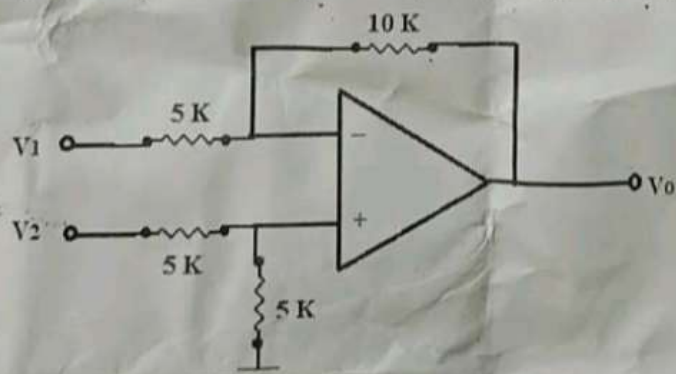


Fig. 4

2(a)	Explain the construction and working operation of N Channel Enhancement type MOSFET with suitable diagram. Also draw its drain and transfer characteristics.	5	CO2	K1, K2
2(b)	What are the properties of an ideal op-amp. Find the output voltage v_o for the Op-Amp circuit given below, where $v_1=10$ volt and $v_2=5$ volt.	5	CO2	K4, K5



4	Explain working of BJT in CB configuration and sketch the input and output characteristics. Also indicate active region, cutoff region and saturation region on the output characteristic curve.	3	CO2
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5	Find I_C and V_{CE} for the circuit of Figure Q. 5, assuming $\beta=100$. Assuming $I_C \cong I_E$.	3	CO3
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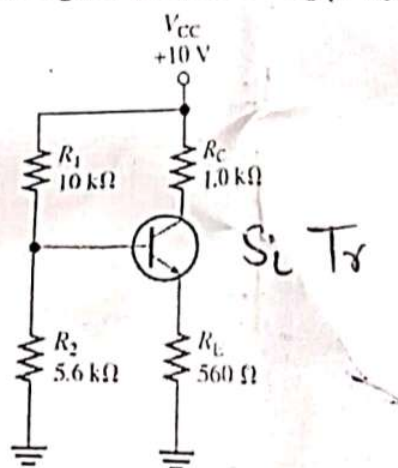


Figure Q. 5

2(a) Explain the input and output characteristic of Common Emitter (CE) amplifier with supporting diagrams. Write the relation between α and β .

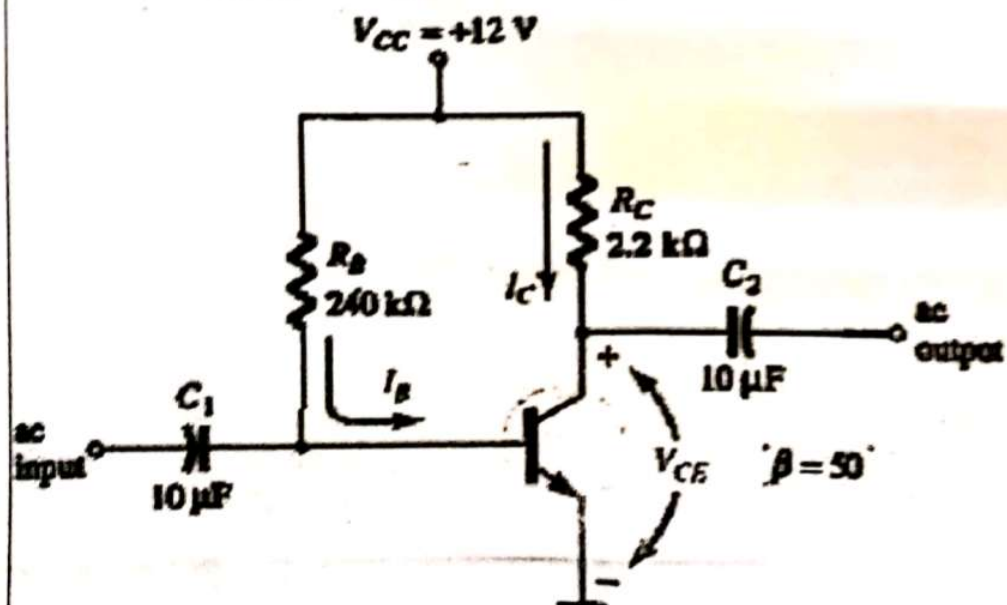
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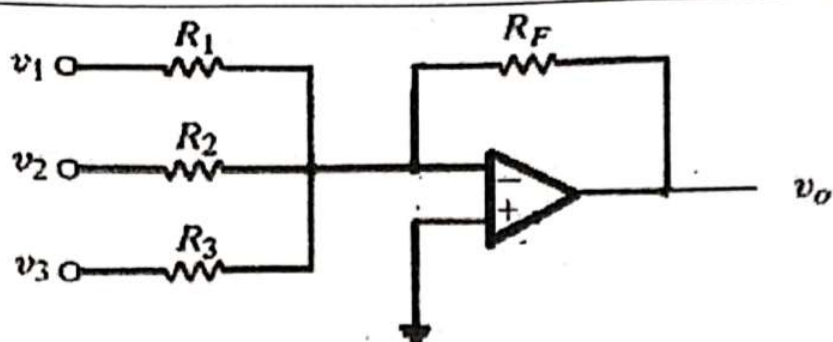
(b) Calculate the I_{BQ} , I_{CQ} and V_{CEQ} for following circuit-

3

2



(c)



For above circuit if $v_1 = x$, $v_2 = 2x$, $v_3 = 3x$ and $R_1 = R_F/2$, $R_2 = R_F/4$ and $R_3 = R_F/2$, Then find the output v_o .

3

2

Q2(a) Given that the $I_C = 2\text{mA}$ and $V_{CE} = 10\text{V}$, determine the value of R_1 and R_C for the below circuit given in Fig.4 . $\beta = 200$

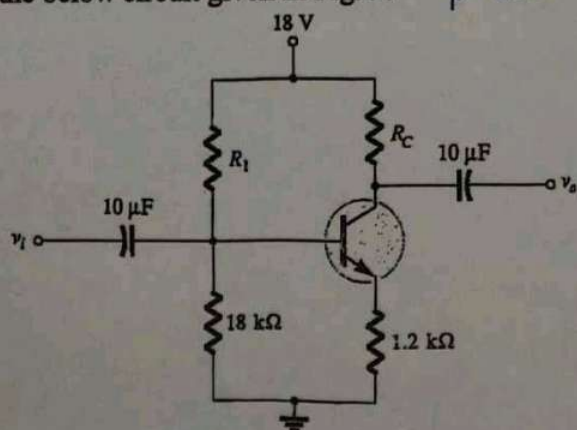


Fig 4

5

CO2

K5

2(a)	Explain the construction and working operation of N Channel Enhancement type MOSFET with suitable diagram. Also draw its drain and transfer characteristics.	5	CO2	K1, K2
2(b)	What are the properties of an ideal op-amp. Find the output voltage v_o for the Op-Amp circuit given below, where $v_1=10$ volt and $v_2=5$ volt.	5	CO2	K4, K5

